

Muown

Muon

$$\text{Diag}(\mathbf{g} / \|\mathbf{R}\|_{\text{row}}) \mathbf{R} =$$

$$\mathbf{W}$$

row-scale

direction

full weight

$$\ell_{\infty}$$

$$\mathcal{S}_{\infty}$$

$$\mathcal{S}_{\infty}$$

$$\mathbf{G}_t \leftarrow \nabla_{\mathbf{W}} \mathcal{L}(\mathbf{W})$$

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compute $\nabla_{\mathbf{g}} \mathcal{L}(\mathbf{g}, \mathbf{R})$ from $\mathbf{G}_t$ use $\nabla_{\mathbf{W}} \mathcal{L}(\mathbf{W})$

$$\mathbf{g} \leftarrow \text{Adam}(\mathbf{g}, \nabla_{\mathbf{g}} \mathcal{L})$$

$$\mathbf{W} \leftarrow \text{Muon}(\mathbf{W}, \nabla_{\mathbf{W}} \mathcal{L})$$

compute $\nabla_{\mathbf{R}} \mathcal{L}(\mathbf{g}, \mathbf{R})$ from $\mathbf{G}_t$

$$\mathbf{R} \leftarrow \text{Muon}(\mathbf{R}, \nabla_{\mathbf{R}} \mathcal{L})$$